3.2 $X=(X_1,X_i,X_m)$

$Y=(Y_1,Y_i,Y_m)$

i indexes examples

$Y_i=f(X_i)=f(X_i;k,b)$

3.3 $\mu=E(X,dim=1)\in R^{1\times m}$

$\sigma^2=E((X-\mu)^2,dim=1)\in R^{1\times m}$

$\hat{X}=\frac{X-\mu}{\sqrt{\sigma^2+\varepsilon}}$, $\varepsilon=10^{-6}$, $\hat{X}=(\hat{X}_1,\hat{X}_i,\hat{X}_m)$

$Y=k\hat{X}+b=k\cdot\hat{X}+b$, $(\cdot$ is ė$)$

$k,b\in R^d$

4. $L=L(Y)\in R$, $\frac{dL}{dY}\Rightarrow\frac{dL}{dk},\frac{dL}{dX},\frac{dL}{db}$

solve:

① $\frac{dL}{dk}=\sum_i^m\frac{dL}{dY_i}\hat{X}_i=sum(\frac{dL}{dY}\cdot\hat{X},dim=2)$ $(\cdot$ is ė$)$

② $\frac{dL}{dX}=s\{u-E(u,dim=1)-E(u\cdot\hat{X},dim=1)\cdot\hat{X}\}$

$(u:=k\cdot\frac{dL}{dY})$, $(\cdot$ is ė$)$

③ $\frac{dL}{db}=\sum_{i=1}^m\frac{dL}{dY_i}=sum(\frac{dL}{dY},dim=2)$

④ $\because\frac{\partial\mu}{\partial x}=\frac{1}{d}(1,\cdots,1)\in R^{1\times d}$

$\frac{\partial\sigma^2}{\partial x}=\frac{2}{d}(x-\mu)^T\in R^{1\times d}$

$\frac{\partial\hat{x}}{\partial x_{direct}}=\frac{1}{\sqrt{\sigma^2+\varepsilon}}I=sI$, $(s:=\frac{1}{\sqrt{\sigma^2+\varepsilon}})$

$\therefore\frac{\partial\hat{x}}{\partial x}=\frac{\partial\hat{x}}{\partial x_{direct}}+\frac{\partial\hat{x}}{\partial\mu}\frac{\partial\mu}{\partial x}+\frac{\partial\hat{x}}{\partial\sigma^2}\frac{\partial\sigma^2}{\partial x}$

$=sI-s\cdot\frac{1}{d}\begin{pmatrix}1\\\vdots\\1\end{pmatrix}(1\cdots1)-\frac{s}{d}\hat{x}(x-\mu)^T$ ($\hat{x}^2$ is elementwise)

$=sI-\frac{s}{d}\begin{pmatrix}1\\\vdots\\1\end{pmatrix}(1\cdots1)-\frac{s}{d}\hat{x}\hat{x}^T$

⑤ $\frac{\partial\gamma}{\partial x}=\frac{\partial\gamma}{\partial\hat{x}}\frac{\partial\hat{x}}{\partial x}=diag(\frac{\partial\gamma}{\partial\hat{x}})\frac{\partial\hat{x}}{\partial x}=diag(k)\frac{\partial\hat{x}}{\partial x}$

⑥ $\because\frac{dL}{dk}=(\frac{\partial L}{\partial\gamma}\frac{\partial\gamma}{\partial k})^T=\frac{dL}{dy}\dot{e}\,\hat{x}$

$\frac{dL}{dx}=(\frac{\partial L}{\partial\gamma}\frac{\partial\gamma}{\partial x})^T=\frac{\partial\hat{x}}{\partial x}\dot{m}(k\dot{e}\frac{dL}{dy})$

$=sI-\frac{s}{d}\begin{pmatrix}1\\\vdots\\1\end{pmatrix}(1\cdots1)u-\frac{s}{d}\hat{x}\hat{x}^Tu$ $(u:=k\dot{e}\frac{dL}{dy})$

$=su-s\frac{(1\cdots1)u}{d}\begin{pmatrix}1\\\vdots\\1\end{pmatrix}-s\frac{(1\cdots1)(\hat{x}\cdot u)}{d}\hat{x}$

$=s(u-E(u)-E(u\cdot\hat{x})\hat{x})$

$\frac{dL}{db}=\frac{dL}{dy}$ (here $E(v):=\frac{1}{d}\sum_{i=1}^d v_i$, $v\in R^d$)

3. $Y=f(X)$, $X,Y\in R^{d\times m}$

m examples with d features

layer normalization

1. $y=f(x)$, $x,y\in R^d$

1.2 $\mu=\frac{1}{d}\sum_{i=1}^d x_i$, $\sigma^2=\frac{1}{d}\sum_{i=1}^d(x_i-\mu)^2$, $\mu,\sigma^2\in R$

$\hat{x}=\frac{x-\mu}{\sqrt{\sigma^2+\varepsilon}}$, $\varepsilon=10^{-6}$

$y=k\hat{x}+b$, $k,b\in R^d$

2. $L=L(y)\in R$, $\frac{dL}{dy}\Rightarrow\frac{dL}{dk},\frac{dL}{dx},\frac{dL}{db}$

$(\frac{dL}{dX}:=(\frac{\partial L}{\partial X}).reshape(X)$, $\forall$ tensor $X)$

solve: (note $\frac{dL}{dy},\frac{dL}{dk},\frac{dL}{dx},\frac{dL}{db}\in R^d$)

① $\frac{\partial\gamma}{\partial k}=\hat{x}$, $\frac{\partial\gamma}{\partial\hat{x}}=k$, $\frac{\partial\gamma}{\partial b}=1$ ($\frac{\partial}{\partial}$ is elementwise)

② $\because\frac{\partial\hat{x}}{\partial\sigma^2}=-\frac{1}{2}(\sigma^2+\varepsilon)^{-\frac{3}{2}}(x-\mu)=-\frac{1}{2}(\sigma^2+\varepsilon)^{-1}\hat{x}$

$\therefore\frac{\partial\gamma}{\partial\sigma^2}=\frac{\partial\gamma}{\partial\hat{x}}\frac{\partial\hat{x}}{\partial\sigma^2}=\frac{\partial\gamma}{\partial\hat{x}}\cdot\frac{\partial\hat{x}}{\partial\sigma^2}=-\frac{1}{2}\frac{k}{\sigma^2+\varepsilon}\hat{x}$ $(\cdot$ is ė$)$

③ $\because\frac{\partial\hat{x}}{\partial\mu_{direct}}=-\frac{1}{\sqrt{\sigma^2+\varepsilon}}\begin{pmatrix}1\\\vdots\\1\end{pmatrix}$, $\frac{\partial\sigma^2}{\partial\mu}=-\frac{2}{d}\sum(x_i-\mu)=0$

(μ_{direct} is μ in def of $\hat{x}$)

$\therefore\frac{\partial\hat{x}}{\partial\mu}=\frac{\partial\hat{x}}{\partial\mu_{direct}}+\frac{\partial\hat{x}}{\partial\sigma^2}\dot{m}\frac{\partial\sigma^2}{\partial\mu}$ ($\dot{m}$ is mat mul)

$=\frac{\partial\hat{x}}{\partial\mu_{direct}}$

$\therefore\frac{\partial\gamma}{\partial\mu}=\frac{\partial\gamma}{\partial\hat{x}}\frac{\partial\hat{x}}{\partial\mu}=\frac{\partial\gamma}{\partial\hat{x}}\dot{e}\frac{\partial\hat{x}}{\partial\mu}=-\frac{k}{\sqrt{\sigma^2+\varepsilon}}$

